

E-Commerce Sales & Website Performance Analysis

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Business Problem & Objectives

Business Problem

The company wants to analyze website traffic, customer behavior, product performance, and marketing effectiveness to improve revenue and customer conversions.

Objectives

- Analyze website traffic and user sessions
- Identify top-performing marketing channels
- Measure website conversion performance
- Understand customer purchasing behavior
- Evaluate product sales and refund trends
- Improve customer engagement and conversion rate

Business Questions

Key Business Questions

1. Which traffic source performs best?
2. Which products generate the highest revenue?
3. Which products should be promoted more aggressively?
4. Which pages attract the highest engagement?
5. Which device generates the most traffic?
6. What factors affect conversion rates?
7. How can conversion rates be improved?
8. How can refund rates be reduced?
9. Which marketing channels underperform?
10. What strategies can improve revenue growth?

Data Set and Tools Used

Datasets Used

- Orders
- Order Items
- Order Item Refunds
- Products
- Website Sessions
- Website Pageviews

Tools Used:

- Excel
- SQL,
- Power BI

Data Cleaning & Preparation

Activities Performed

- Imported six datasets
- Checked missing values
- Removed duplicate records
- Validated relationships between tables
- Converted date fields into proper format
- Verified primary and foreign keys
- Standardized data types

SQL Analysis Overview

SQL Techniques Used

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- COUNT()
- SUM()
- INNER JOIN
- Aggregate Functions

Analysis Performed

- Website traffic analysis
- Product performance analysis
- Revenue analysis
- Refund analysis
- Marketing channel analysis

SQL Analysis Findings

Analysis 1: Website Traffic Performance

Query Objective

Identify the traffic sources generating the highest customer activity.

Findings

- gsearch generated the highest website traffic.
- Social channels contributed the least traffic.
- Search-based traffic dominated customer acquisition.

Business Insight

Search traffic, particularly gsearch, is the strongest source of website traffic, making it an important channel for customer acquisition and marketing focus.

Analysis 2: Order Performance

Query Objective

Calculate total orders generated across the website.

Findings

- Total Orders: 32.31K
- Orders increased significantly over time.

Business Insight

Growing order volume indicates increasing customer demand and business growth.

Analysis 3: Revenue Performance

Query Objective

Calculate total revenue generated from customer purchases.

Findings

- Total Revenue: \$1.94M
- Revenue peaked during 2014-2015.

Business Insight

The business experienced strong sales growth over time.

Analysis 4: Product Performance

Query Objective

Identify top-performing products based on revenue.

Findings

Top Revenue Generating Product:

- The Original Mr. Fuzzy

Other products are performing less.

- The Forever Love Bear
- The Hudson River Mini Bear
- The Birthday Sugar Panda

Business Insight

One product generate most of the company's revenue.

Analysis 5: Refund Analysis

Query Objective

Analyze refund activity across products.

Findings

- Refunds were observed across multiple products.
- Certain products experienced more refund activity than others.

Business Insight

Refunds directly impact profitability and customer satisfaction.

Key Insights Generated Through SQL

- Highest performing traffic source identified
- Total orders calculated
- Revenue trends analyzed
- Top-selling products identified
- Refund patterns evaluated
- Customer behavior insights extracted

Analysis and Visualization in Power BI

DAX Measures Created

Measure 1: Conversion Rate

Orders ÷ Website Sessions × 100

Result: **6.83%**

Measure 2: Average Order Value

Revenue ÷ Orders

Result: **\$59.99**

Measure 3: Refund Rate

Formula: Refunded Orders ÷ Total Orders ×
100

Result: **4.32%**

Dashboard 1

KPI Cards

- Website Sessions = 472.87K
- Total Orders = 32.31K
- Revenue = \$1.94M
- Average Order Value = \$59.99
- Conversion Rate = 6.83%

472.87K

Total Website Sessions

32.31K

Total Orders

1.94M

Total Revenue

59.99

Average Order Value

6.83

Conversion Rate

472.87K

Total Website Sessions

32.31K

Total Orders

1.94M

Total Revenue

59.99

Average Order Value

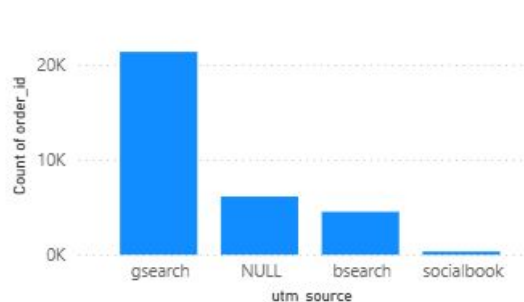
6.83

Conversion Rate

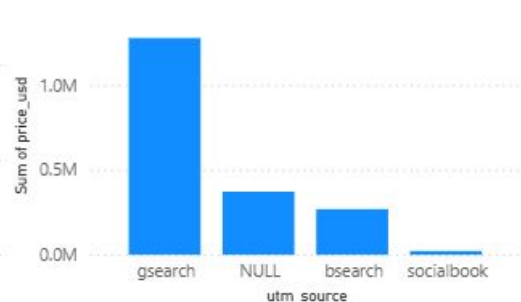
Revenue Trend Over Time



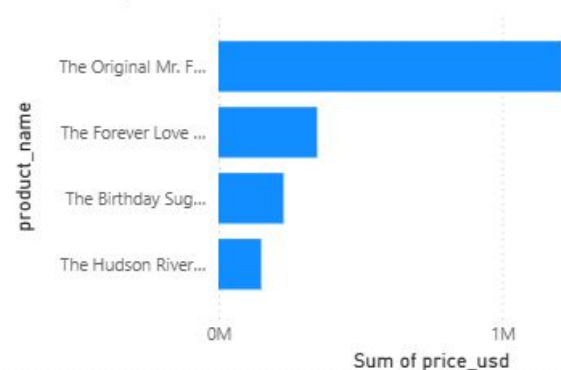
Orders by Traffic Source



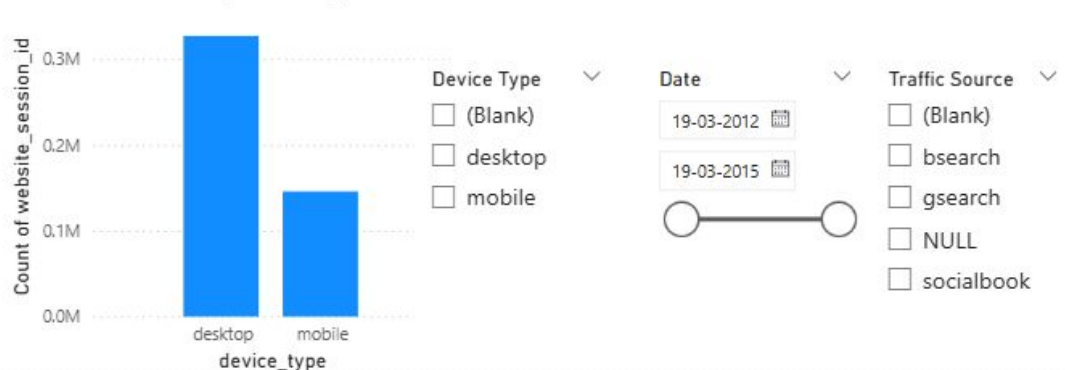
Revenue by Traffic source



Revenue by Product



Website Sessions By Device Type

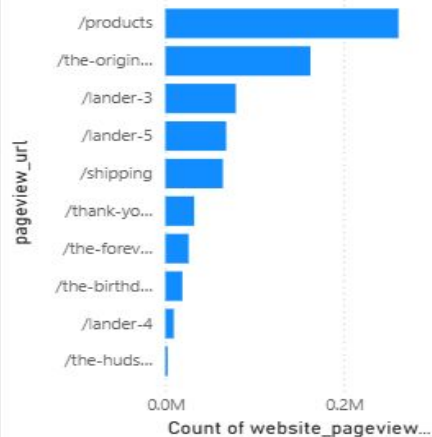


Key Findings

- The website generated 472.87K sessions and 32.31K orders, resulting in a 6.83% conversion rate.
- Total revenue reached \$1.94M, with an average order value of \$59.99.
- Desktop generated substantially more website sessions than mobile, indicating stronger desktop traffic.
- Search traffic, particularly gsearch, generated the highest number of orders and the largest revenue contribution.
- Revenue increased substantially over the later years, with the strongest activity visible around 2014–2015.
- The Original Mr. Fuzzy is the leading revenue-generating product, contributing significantly more revenue than the other products.

Dashboard 2

Most Visited Pages



44.76%

Bounce Rate

6.83

Conversion Rate

472.87K

Total Website Sessions

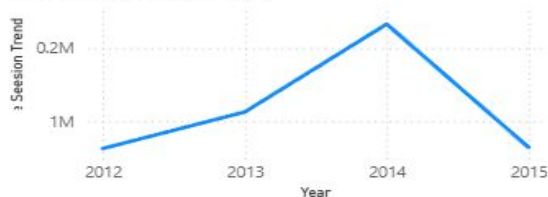
59.99

Average Order Value

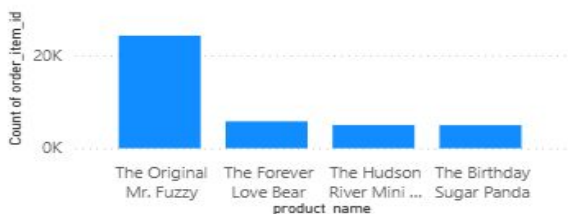
4.32

Refund Rate

Website Sessions Trend



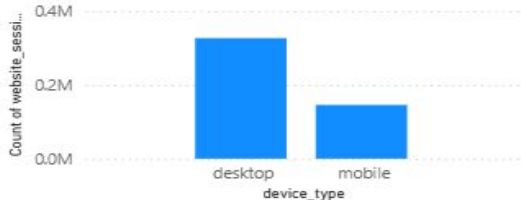
Units Sold By Product



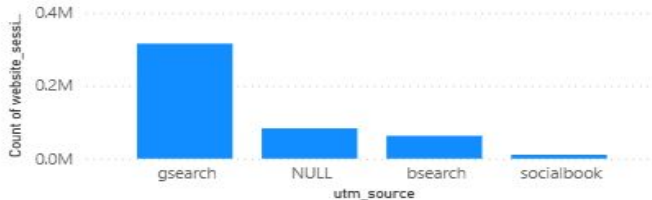
Refund By Products



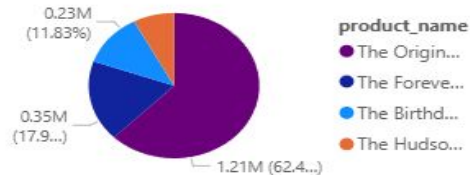
Website Sessions By Device Type



Website Sessions By Traffic Source



Revenue Contribution By Product



Website Insights

Findings

- **Product pages** received the highest number of page views, indicating **strong customer interest** in product exploration.
- **Search traffic** generated the largest share of website sessions, with gsearch leading the traffic sources.
- Website sessions peaked in **2014** before declining in 2015.
- **Desktop** users generated substantially more sessions than mobile users.
- **The bounce rate was 44.76%**, indicating that a significant portion of sessions ended after limited page interaction.

Insight

Customers show **strong interest** in exploring product pages, while the high bounce rate highlights an opportunity to improve engagement and navigation.

Product Insights

Findings

- **The Original Mr. Fuzzy** generated the highest number of units sold.
- Product revenue is concentrated among a small number of products, with The Original Mr. Fuzzy contributing the **largest share**.
- Refund activity varies across products, with The Original Mr. Fuzzy recording the **highest number of refunds**.

- The overall refund rate is **4.32%**.

Insight

This one Product products drives a substantial share of sales and revenue, making these products important for inventory, marketing, and retention strategies.

Customer Insights

Findings

- **Desktop users** generated significantly more website sessions than mobile users.
- **gsearch** was the **dominant traffic source**, generating the highest number of sessions.
- The overall conversion rate was 6.83%.
- Average Order Value was \$59.99.

- Customer behavior differs across devices and traffic sources, providing opportunities for **targeted optimization**.

Insight

Desktop and search traffic are the strongest contributors to customer activity, while device- and source-specific optimization can help improve conversion performance.

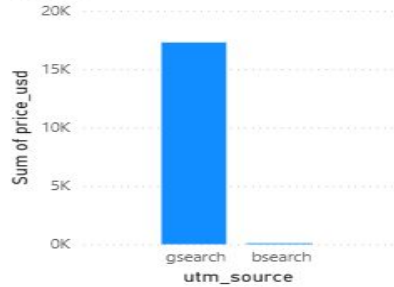
Dashboard 3

This dashboard satisfies the project requirement of using a filtered dataset.

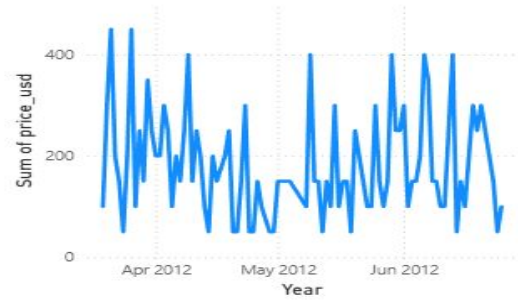
17.35K

Total High Value Revenue

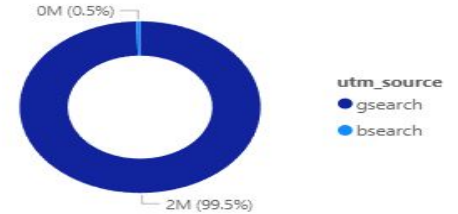
High Value Revenue by traffic Source



High Value Revenue Trend



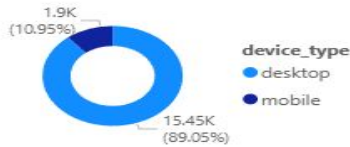
High Value Sessions By Traffic Source



49.99

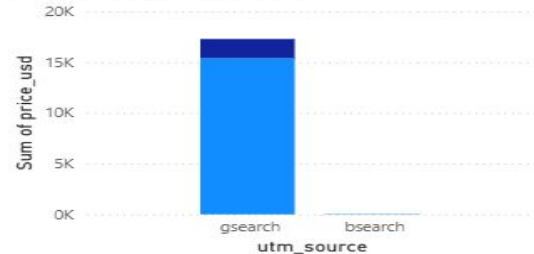
Average High Order Value

High Value Revenue By Device Type

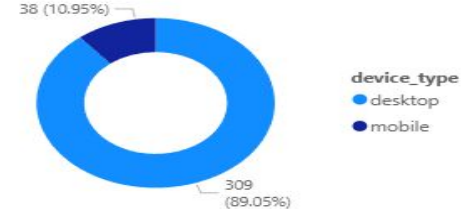


High Value Revenue By Traffic Source And Device

device_type ● desktop ● mobile



High Value Sessions By Device Type



347

High Value Sessions

Findings

- The SQL filter isolated 347 high-value converted sessions.
- Total revenue from the filtered high-value segment was \$17.35K.
- The average order value within the filtered segment was \$49.99.
- gsearch generated 99.5% of the high-value
- Desktop users accounted for 89.05% of high-value sessions, compared with 10.95% from mobile.
- Desktop also generated the majority of high-value revenue, contributing approximately \$15.45K.

Insight

High-value conversions are heavily concentrated in gsearch and desktop users, suggesting that these segments are the strongest targets for high-value customer acquisition and retention.

Business Insights

Which traffic source performs best and why?

Answer: gsearch

Supporting Evidence

- Highest website sessions
- Highest orders
- Strongest conversion performance

Insight

Search traffic attracts customers with higher purchase intent.

Which product should the company promote more aggressively?

Answer:

- The Original Mr. Fuzzy

Supporting Evidence

This products generated the highest revenue.

What factors affect conversion rates?

Supporting Evidence

- Mobile conversion rate (3.09%) is lower than overall conversion rate (6.83%).
- Traffic quality varies across channels.

Insight

Device experience and marketing channel quality influence conversions.

Suggest one strategy to improve customer conversions.

Recommendation

Improve mobile website experience by:

- Faster loading pages
- Simplified checkout process
- Better navigation

Suggest one strategy to reduce refund rates.

Recommendation

Improve product information and quality monitoring.

This will reduce customer dissatisfaction and unnecessary returns.

Overall Project Insights

Major Insights

1. Website generated over 472K sessions.
2. Revenue reached approximately \$1.94M.
3. Conversion rate achieved 6.83%.
4. gsearch is the highest-performing traffic source.
5. Desktop users generate the most traffic.
6. Product pages attract the highest engagement.
7. A few products generate most of the revenue.
8. Mobile conversion performance is lower than overall conversion performance.
9. Revenue increased significantly over time.
10. Refund activity affects profitability.

Business Recommendations

Marketing Strategy

- **Increase investment in gsearch campaigns**, which generated the highest order volume and revenue.
- **Prioritize high-value desktop audiences**, which represented **89.05% of high-value sessions** in the SQL-filtered analysis.
- **Evaluate bsearch and other lower-performing channels** before increasing marketing spend.
- **Use traffic-source and device-level segmentation** to create more targeted campaigns.
- **Monitor campaign performance using both revenue and conversion rate**, rather than traffic volume alone.

Product Strategy

- **Prioritize The Original Mr. Fuzzy**, the leading product in both sales volume and revenue contribution.
- **Maintain adequate inventory for high-performing products** to avoid losing sales during periods of strong demand.
- **Bundle top-selling products with complementary products** to encourage larger purchases.
- **Review lower-performing products** and consider better positioning, promotions, or revised product messaging.

Customer Experience

- **Improve the mobile shopping experience**, particularly navigation and checkout, because desktop users dominate both overall and high-value activity.
- **Reduce the 44.76% bounce rate** by improving landing pages, navigation, page speed, and calls-to-action.
- **Optimize product pages**, since product-related pages receive the highest page-view activity.
- **Create a smoother path from product discovery to checkout** to convert more of the existing website traffic.
- **Use customer behavior by device and traffic source to personalize the shopping experience.**

Refund Reduction

- **Investigate products with the highest refund volumes**, especially The Original Mr. Fuzzy.
- **Improve product descriptions, specifications, images, and sizing/details** to reduce expectation gaps.
- **Strengthen quality-control checks for frequently refunded products.**
- **Monitor refund rate regularly**, currently **4.32%**, and identify products or periods where it increases.
- **Analyze the reason for refunds**, where additional data is available, and use the findings to improve products and fulfillment.

High-Value Customer Strategy

- **Prioritize gsearch for high-value customer acquisition**, as it generated **99.5% of high-value sessions** in the filtered analysis.
- **Develop desktop-focused campaigns for high-value customers**, since desktop represented **89.05% of high-value sessions**.
- **Create separate strategies for high-value and general visitors**, rather than treating all website traffic equally.
- **Track high-value revenue by traffic source and device continuously** to identify the strongest customer segments.
- **Develop retention campaigns for customers acquired through the strongest high-value segments**.

Revenue Growth

- **Focus resources on the products and traffic sources that contribute the greatest revenue.**
- **Use the \$59.99 overall average order value as a benchmark** and test bundles or cross-selling strategies to increase it.
- **Investigate the revenue spikes shown in the time trend** to identify campaigns, products, or seasonal factors that drove those increases.

Thank You